Towards Fluid Networking: Citizen-Co-Created Low-Height Distributed MIMO Access Networks

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***Abstract*—This paper studies low-height distributed-multiple-input multiple-output (MIMO) access networks with repositionable access points in an outdoor urban setting. Using the `COCOON-Sionna` framework, the workflow generates wall-mounted candidate access points (APs), simulates pedestrian and bicycle mobility, computes path-based channel state information (CSI), and compares a central rooftop proxy, a fixed distributed deployment, and three relocation-capable distributed deployments over a common scenario. The article formalizes the signal-to-interference-plus-noise ratio (SINR)- and ergodic sum rate (ESR)-based evaluation model, a history-aware K-nearest relocation heuristic that scores exponentially aged best-user deployed-AP-to-user equipment (UE) SINR samples, a nearest-user UE-to-UE proxy-CSI heuristic that evaluates a proxy-ESR objective from wideband proxy channels, and a nearest-user proxy-power heuristic that ranks candidate subsets from stored UE-to-UE link-power measurements scaled to the AP transmit-power budget. In the Rabot scenario, the proxy-ESR controller is the strongest strategy: it reaches a mean ESR of 133.1 bit/s/Hz, improves the 10th-percentile ESR by 88.8 % over the fixed distributed baseline, and reduces the 0 dB outage fraction from 4.41 % to 1.87 %. The power-only proxy controller is second-best in radio performance but requires more motion, whereas the historical local-CSI controller improves average ESR over the static baseline while substantially degrading the lower-tail SINR. The paper also documents the postprocessing workflow that produces geometry, performance, and relocation-cost figures from reproducible simulation outputs.**

***Index Terms*—cell-free systems, distributed MIMO, low-height access points, access-point placement, urban mobility, Sionna RT**

I. Introduction

The Cocoon project targets a wireless access paradigm in which the network adapts to the fluidity of the city rather than assuming a static infrastructure and a stationary demand profile. The motivating hypothesis from the project proposal is that a practical cell-free (CF) network becomes more feasible when low-height infrastructure can be co-created, repositioned, and activated only when the measured channel conditions justify it. In that view, the citizens are not only traffic sources but also anchors for sensing, mounting, and reconfiguring the surrounding network.

This article narrows that broader vision to a simulation-based study of low-height infrastructure adaptation. The implemented framework covers wall-mounted candidate access points (APs), outdoor user mobility, path-based channel state information (CSI) extraction, and repeated placement decisions over relocation windows in an urban scene. This scope enables a reproducible comparison of geometry, radio-performance, and relocation-cost metrics for alternative infrastructure strategies.

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Fig. 1: Conceptual illustration of the Cocoon low-height access-network vision.

The main contributions of the present study are:

- a concise technical framing of Cocoon as a low-height, movable-AP, citizen-co-created access-network concept,
- a reproducible description of the Rabot outdoor simulation pipeline implemented in `COCOON-Sionna`, and
- a common analysis workflow for geometry, signal-to-interference-plus-noise ratio (SINR), ergodic sum rate (ESR), and relocation-cost results derived from exported simulation artifacts.

II. Scope Within Cocoon

The full Cocoon proposal spans citizen interaction with the wireless network, channel assessment, architecture design, and proof-of-concept validation. The present article addresses the simulation-facing part of that program, with emphasis on dynamic channel-aware adaptation, low-height AP/contact service point (CSP) placement, and the algorithmic consequences of allowing physical repositioning.

The main modeling simplification is that co-creation is represented here through an abstract movable infrastructure layer rather than through a user study or a deployment trial. Candidate mounting locations stand in for positions that could later be offered to citizens, and the placement scheduler stands

in for the future tool chain that would notify participants where the next best CSP constellation should be installed.

Accordingly, the scope of the present article is:

- the paper validates a simulation methodology and a comparison workflow, not the social feasibility of citizen participation,
- the paper studies a Rabot outdoor scenario, not yet a multi-site or multi-season campaign,
- the paper compares placement strategies over a common CSI-derived objective, not yet a full end-to-end network stack.

III. System Model

We consider a downlink distributed MIMO (D-MIMO) system with active APs $\mathcal{A} = \{1, \ldots, M\}$ and scheduled user equipments (UEs) $\mathcal{U} = \{1, \ldots, K\}$. Let N_m denote the number of transmit ports at AP m, let $N_{r,k}$ be the number of receive ports at UE k, and let $f \in \{1, \ldots, F\}$ index the sampled CFR tones over the $100\,\mathrm{MHz}$ band. The per-AP channel block on tone f is

$$\mathbf{H}_{k,m}[f] \in \mathbb{C}^{N_{r,k} \times N_m}, \tag{1}$$

and the full distributed channel observed by UE k is

$$\mathbf{H}_k[f] = \left[\mathbf{H}_{k,1}[f]\ \mathbf{H}_{k,2}[f]\ \cdots\ \mathbf{H}_{k,M}[f]\right]. \tag{2}$$

The antenna topology follows the `PlanarArray` abstraction used by the simulator. Each node is modeled as a uniform planar array (UPA) with N^{row} rows and N^{col} columns, so that the number of polarization-agnostic elements is $N^{\mathrm{row}} N^{\mathrm{col}}$. The implementation uses the same element radiation pattern, polarization setting, and inter-element spacing on transmit and receive arrays. More precisely, the number of ports is

$$N = N^{\mathrm{row}} N^{\mathrm{col}} N^{\mathrm{pol}}, \tag{3}$$

where N^{pol} is the number of active polarization ports implied by the configured polarization string. In the Rabot configuration, the APs use a 2×3 UPA with `VH` dual polarization, which yields $N_m = 2 \cdot 3 \cdot 2 = 12$ transmit ports per AP. The UEs use a 1×1 planar array with the same `VH` dual-polarized element model, which yields $N_{r,k} = 1 \cdot 1 \cdot 2 = 2$ receive ports per user device. The vertical and horizontal element spacings are both set to 0.5λ, so the array factor corresponds to half-wavelength sampling in both planar directions.

The element radiation pattern is the 3rd generation partnership project (3GPP) `tr38901` pattern provided by Sionna. Thus, the simulated channel is not isotropic at the element level: every array response combines the `tr38901` element directivity with the UPA array factor determined by the node orientation, row-column topology, and the propagation geometry. This matters for the present study because the wall-mounted distributed APs and the rooftop central proxy are evaluated with the same underlying element pattern while differing in their locations, orientations, and distributed versus centralized array geometry.

For the central reference, the implementation collapses the distributed element budget into one normalized co-located rooftop array. If each distributed AP has $N_{\mathrm{AP}}^{\mathrm{row}} N_{\mathrm{AP}}^{\mathrm{col}}$ physical element locations and the distributed budget contains M_{dist} APs, then the central proxy uses

$$N_{\mathrm{cent}}^{\mathrm{elem}} = M_{\mathrm{dist}} N_{\mathrm{AP}}^{\mathrm{row}} N_{\mathrm{AP}}^{\mathrm{col}} \tag{4}$$

element locations, reshaped into a factorized UPA with integer row and column counts. In the Rabot configuration, $M_{\mathrm{dist}} = 6$ and the distributed hardware uses 2×3 elements per AP, so the normalized co-located rooftop proxy becomes a 6×6 UPA. With the same `VH` dual polarization, this yields $6 \cdot 6 \cdot 2 = 72$ transmit ports. The central transmit-power budget is scaled to the sum power of the six distributed APs. This keeps the central reference aligned with the distributed element and power budgets, while still remaining a rooftop proxy rather than a full placement model for a true co-located base-station site.

Given a receive combiner $\mathbf{u}_k[f]$, the effective channel becomes

$$\mathbf{h}_k^{\mathsf{H}}[f] = \mathbf{u}_k^{\mathsf{H}}[f] \mathbf{H}_k[f]. \tag{5}$$

Stacking all users yields the effective distributed channel matrix $\mathbf{H}_{\mathrm{eff}}[f]$. The current pipeline applies a least-squares pseudo-inverse zero-forcing (ZF) precoder independently on every sampled tone, followed by per-stream power allocation under a total AP power budget.

For UE k on tone f, the instantaneous downlink SINR is evaluated as

$$\mathrm{SINR}_k^{\mathrm{DL}}[f] = \frac{p_k \left|\mathbf{h}_k^{\mathsf{H}}[f] \mathbf{f}_k[f]\right|^2}{\sum_{i \in \mathcal{U},\, i \neq k} p_i \left|\mathbf{h}_k^{\mathsf{H}}[f] \mathbf{f}_i[f]\right|^2 + \sigma^2}, \tag{6}$$

where $\mathbf{f}_i$ is the precoding vector for stream i, p_i its allocated power, and σ^2 the thermal-noise term. The implementation keeps the interference term explicitly and exports desired-signal, interference, and noise power terms for each evaluated user snapshot.

The implementation evaluates this expression on all sampled tones and stores a scalar wideband SINR per user by mapping the tone-averaged spectral efficiency back to a Shannon-equivalent SINR. The ESR used later in the results section is derived from the same full-band per-snapshot SINR samples. For snapshot n with scheduled-user set $\mathcal{U}_n$, the postprocessing layer computes

$$\mathrm{ESR}_n = \sum_{k \in \mathcal{U}_n} \frac{1}{F} \sum_{f=1}^{F} \log_2\left(1 + \mathrm{SINR}_{k,n}^{\mathrm{DL}}[f]\right). \tag{7}$$

These per-snapshot SINR values form the basis of the trajectory-level SINR and ESR statistics reported later in the article.

A. Placement Strategies

The placement layer compares five strategies under a common AP budget:

- `central_massive_mimo`: one rooftop proxy AP is placed on the building whose rooftop representative point is nearest the area center; the proxy is oriented toward

the area center. Its antenna topology is the normalized co-located array obtained by aggregating the distributed element budget into one rooftop UPA. This is a normalized central reference, not a full single co-located-array placement model.

- `distributed_fixed`: a distributed set of wall-mounted candidate positions is sampled once and kept fixed over the full trajectory.
- `distributed_movable`: optimization 1. The same number of distributed APs is reselected at each relocation window from the wall-mounted candidate pool with a history-aware local-CSI rule.
- `distributed_movable_optimization_2`: optimization 2. The same number of distributed APs is reselected at each relocation window from the wall-mounted candidate pool with a current-window nearest-user UE-to-UE proxy-CSI rule and a proxy-ESR objective.
- `distributed_movable_optimization_3`: optimization 3. The same number of distributed APs is reselected at each relocation window from the wall-mounted candidate pool with the same nearest-user proxy association as optimization 2, but with a proxy-average-power objective based on $|\mathrm{CSI}|^2$.

Let $\mathcal{C}$ denote the wall-mounted candidate pool and let $\mathcal{S}_w \subseteq \mathcal{C}$ denote the selected movable-AP subset in relocation window w. In the Rabot scenario studied here, $|\mathcal{S}_w| = N_{\mathrm{mov}} = 6$ and no always-on distributed APs are used, so the distributed comparison isolates the effect of changing the movable subset over time.

B. Relocation Heuristic

At an intuitive level, the relocation mechanism repeats the same observe-score-relocate loop at every relocation instant. Optimization 1 evaluates already simulated deployed-AP-to-UE samples for the subsets that have been queried so far and discounts older window segments exponentially. Optimization 2 and optimization 3 follow a different practical assumption: the AP-to-UE channel of a dormant candidate position is not known before an AP is actually moved there. Consequently, these two rules do not optimize directly over candidate-position AP-to-UE CSI. Instead, each candidate site is associated with the closest user within a distance threshold. Optimization 2 reuses the measured UE-to-UE wideband channel of that nearby user as a proxy and ranks the subset by proxy ESR. Optimization 3 uses the corresponding stored UE-to-UE link-power samples, scaled to the AP transmit-power budget, and ranks the subset by proxy average power. Once the best subset has been chosen, the movable APs are relocated to those candidate positions and the actual deployed AP-to-UE CSI is recomputed for storage, postprocessing, and final reporting.

The movable distributed strategy is implemented as a windowed receding-horizon controller. Let $\{t_n\}_{n=0}^{N_t-1}$ denote the simulated snapshot times and let T_{rel} denote the relocation interval. The trajectory is partitioned into disjoint decision windows

$$\mathcal{W}_w = \left\{ n \in \{0, \ldots, N_t - 1\} : \left\lfloor \frac{t_n - t_0}{T_{\mathrm{rel}}} \right\rfloor = w \right\}. \tag{8}$$

At the beginning of each window w, the controller selects a subset $\mathcal{S}_w \subseteq \mathcal{C}$ with cardinality $|\mathcal{S}_w| = N_{\mathrm{mov}}$, where N_{mov} is the movable-AP budget. The fixed distributed mode uses one time-invariant subset, whereas the movable mode recomputes $\mathcal{S}_w$ at every relocation instant.

a) Optimization 1: historical K-nearest local-CSI controller.: The first online decision rule is intentionally local and does not optimize a full radio map. For a candidate site $c \in \mathcal{C}$ with horizontal position $\mathbf{x}_c \in \mathbb{R}^2$, define the local neighborhood $\mathcal{N}_K(c, w)$ as the set of the K nearest UE-snapshot pairs in the horizontal plane within window w,

$$\mathcal{N}_K(c, w) = \mathrm{KNN}_K(\mathbf{x}_c; \{\mathbf{r}_{u,n} : u \in \mathcal{U}, n \in \mathcal{W}_w\}), \tag{9}$$

where $\mathbf{r}_{u,n}$ denotes the horizontal position of UE u at time index n. For a subset $\mathcal{S} \subseteq \mathcal{C}$, the support region used by the controller is the union

$$\mathcal{N}_K(\mathcal{S}, w) = \bigcup_{c \in \mathcal{S}} \mathcal{N}_K(c, w). \tag{10}$$

In the Rabot configuration, $K = 10$, so each candidate contributes at most ten local UE-snapshot pairs per window. This restriction makes the relocation rule sensitive to the CSI observed near the candidate sites rather than to globally averaged scene statistics. In other words, user positions are used to construct the local sampling mask, but they do not enter the score directly.

For every queried subset $\mathcal{S}$, the implementation computes the corresponding deployed AP-to-UE CSI/SINR segment over the current window and stores it in a per-subset cache. Thus, the relocation heuristic is driven by channel-response information associated with the candidate subset under evaluation, not by geometry alone. If $\gamma_{u,n}^{\mathrm{dB}}(\mathcal{S})$ denotes the stored best-link SINR sample in dB produced when subset $\mathcal{S}$ is active, the historical record available when deciding window w is

$$\mathcal{H}_w(\mathcal{S}) = \left\{ (t_n, \gamma_{u,n}^{\mathrm{dB}}(\mathcal{S})) : u \in \mathcal{U}, n \in \mathcal{W}_j, j = 0, \ldots, w \right\}. \tag{11}$$

When the same subset is revisited later, previously computed window segments are reused and only the new current-window segment is appended. Thus, the historical objective remains consistent while avoiding repeated ray-tracing evaluations for past windows. Within optimization 1, the relocation score is formed from these deployed AP-to-UE samples only; UE-to-UE CSI does not enter this objective, and AP-to-AP CSI is exported for analysis rather than for online relocation control.

Historical CSI is discounted through exponential aging. If a sample was observed at time t_n and the relocation decision for window w is taken at t_w^{end}, the associated weight is

$$\omega_n^{(w)} = \exp\left[-\lambda \left(t_w^{\mathrm{end}} - t_n\right)\right], \tag{12}$$

with λ the configured decay rate. Hence, newer samples always dominate older ones: a sample measured at the decision

time keeps weight 1, a sample that is one relocation window old keeps weight $\exp(-\lambda T_{\mathrm{rel}})$, and the weight keeps halving every additional relocation window when $\lambda = \ln(2)/T_{\mathrm{rel}}$, as in the Rabot configuration. The local control objective is then the weighted empirical 10th percentile of the SINR samples retained by the local neighborhoods across all observed windows,

$$J_w^{(1)}(\mathcal{S}) = Q_{0.1}^{\omega}\left(\left\{\gamma_{u,n}^{\mathrm{dB}}(\mathcal{S}) : (u,n) \in \bigcup_{j=0}^{w} \mathcal{N}_K(\mathcal{S}, j)\right\}\right), \tag{13}$$

where $Q_{0.1}^{\omega}(\cdot)$ denotes the weighted 10th-percentile operator induced by $\omega_n^{(w)}$. The use of a lower-tail percentile, rather than a mean, makes the controller deliberately conservative with respect to persistently weak local links.

b) Optimization 2: current-window nearest-user proxy-CSI controller.: The second online rule removes the K-nearest construction and does not use historical samples. Let d_{th} denote the configured distance threshold. For each candidate site $c \in \mathcal{C}$ and snapshot $n \in \mathcal{W}_w$, define the anchor user

$$\ell(c,n) = \arg\min_{u \in \mathcal{U}} \|\mathbf{r}_{u,n} - \mathbf{x}_c\|_2, \tag{14}$$

and the validity indicator

$$\delta(c,n) = \mathbf{1}\left\{\left\|\mathbf{r}_{\ell(c,n),n} - \mathbf{x}_c\right\|_2 \le d_{\mathrm{th}}\right\}. \tag{15}$$

The dormant candidate-position AP-to-UE channel is not available before relocation. Therefore, optimization 2 replaces it by a proxy channel extracted from the measured UE-to-UE CSI. If $h_{u,v,n}^{\mathrm{UU}}[f]$ denotes the sampled UE-to-UE coefficient from transmitter-user v to receiver-user u at snapshot n and tone f, the proxy coefficient used for candidate c is

$$\tilde{h}_{u,c,n}[f] = \begin{cases} h_{u,\ell(c,n),n}^{\mathrm{UU}}[f], & \delta(c,n) = 1,\ u \ne \ell(c,n), \\ h_{v^\star,\ell(c,n),n}^{\mathrm{UU}}[f], & \delta(c,n) = 1,\ u = \ell(c,n), \\ 0, & \delta(c,n) = 0, \end{cases} \tag{16}$$

where $v^\star = \arg\max_{v \in \mathcal{U}} \max_f |h_{v,\ell(c,n),n}^{\mathrm{UU}}[f]|$ supplies a surrogate self-link because the diagonal user-to-itself measurement is not observable. For a candidate subset $\mathcal{S}$, the proxy channel matrix $\tilde{\mathbf{H}}_n[f](\mathcal{S})$ is formed by stacking the columns $\tilde{h}_{u,c,n}[f]$ for all $c \in \mathcal{S}$. The same ZF procedure as in the deployed AP case is then applied on every sampled tone, yielding a proxy SINR $\tilde{\gamma}_{u,n}[f](\mathcal{S})$ and the current-window proxy-ESR objective

$$J_w^{(2)}(\mathcal{S}) = \frac{1}{|\mathcal{W}_w| F} \sum_{n \in \mathcal{W}_w} \sum_{f=1}^{F} \sum_{u \in \mathcal{U}} \log_2(1 + \tilde{\gamma}_{u,n}[f](\mathcal{S})). \tag{17}$$

Consequently, optimization 2 uses measured UE-to-UE proxy CSI to rank candidate subsets, while avoiding the unrealistic assumption that dormant candidate-position AP-to-UE channels are already known. In the software implementation, this is the preferred path whenever a consistent wideband UE-to-UE CFR export is available. If the full complex proxy tensor is unavailable or cannot be decoded reliably, the code falls back to a power-based proxy-ESR computation built from the stored UE-to-UE link-power tensor so that the optimization can still be executed.

c) Optimization 3: current-window nearest-user proxy-power controller.: The third online rule reuses the same nearest-user anchor $\ell(c,n)$ and validity indicator $\delta(c,n)$ as optimization 2, but it does not use the complex proxy coefficient directly. Instead, it operates on the stored UE-to-UE link-power tensor. Let $\rho_{a,u,n}^{\mathrm{UU}}$ denote the ordered link-power entry returned by the simulator between anchor-user index a and user index u at snapshot n. The implementation forms the per-candidate proxy-power row

$$\bar{\rho}_{u,c,n} = \begin{cases} \rho_{\ell(c,n),u,n}^{\mathrm{UU}}, & \delta(c,n) = 1,\ u \ne \ell(c,n), \\ \max_{v \in \mathcal{U},\, v \ne \ell(c,n)} \rho_{\ell(c,n),v,n}^{\mathrm{UU}}, & \delta(c,n) = 1,\ u = \ell(c,n), \\ 0, & \delta(c,n) = 0, \end{cases} \tag{18}$$

and then rescales that row to the distributed-AP transmit-power budget according to

$$\tilde{\rho}_{u,c,n} = \frac{P_{\mathrm{AP}}}{P_{\mathrm{UE}}} \bar{\rho}_{u,c,n}. \tag{19}$$

The optimization-3 objective is then

$$J_w^{(3)}(\mathcal{S}) = \frac{1}{|\mathcal{W}_w|} \sum_{n \in \mathcal{W}_w} \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \sum_{c \in \mathcal{S}} \tilde{\rho}_{u,c,n}. \tag{20}$$

Optimization 3 is therefore the power-only counterpart of optimization 2: it uses the same nearest-user proxy association and the same self-link substitution rule, but it requires only stored proxy link power rather than the complex proxy channel.

d) Greedy subset construction.: All three optimization rules use the same forward-selection search. Starting from $\mathcal{S}_w^{(0)} = \varnothing$, the controller adds one candidate at a time according to

$$c_r^\star = \arg\max_{c \in \mathcal{C} \setminus \mathcal{S}_w^{(r-1)}} J_w^{(q)}\left(\mathcal{S}_w^{(r-1)} \cup \{c\}\right), \tag{21}$$

$$q \in \{1,2,3\}, \qquad \mathcal{S}_w^{(r)} = \mathcal{S}_w^{(r-1)} \cup \{c_r^\star\}, \qquad r = 1, \ldots, N_{\mathrm{mov}}, \tag{22}$$

and finally sets $\mathcal{S}_w = \mathcal{S}_w^{(N_{\mathrm{mov}})}$. Consequently, the current implementation is a forward-selection heuristic rather than an exhaustive combinatorial optimizer.

After either optimization rule has chosen the subset, the selected geometric candidate sites are converted into physical relocations of labeled movable APs. Let $\mathbf{p}_m^{(w-1)}$ denote the previous-window position of movable AP m and let $\mathbf{q}_j^{(w)}$ denote the coordinates of the candidates in the newly selected subset. The controller solves the linear assignment problem

$$\pi_w^\star = \arg\min_{\pi \in \Pi_{N_{\mathrm{mov}}}} \sum_{m=1}^{N_{\mathrm{mov}}} \left\|\mathbf{p}_m^{(w-1)} - \mathbf{q}_{\pi(m)}^{(w)}\right\|_2^2, \tag{23}$$

where $\Pi_{N_{\mathrm{mov}}}$ is the permutation set on $\{1, \ldots, N_{\mathrm{mov}}\}$. This minimum-displacement matching preserves AP identities across windows and yields schedule traces that correspond to physically meaningful AP trajectories instead of arbitrary candidate relabelings.

Once all windows have been simulated, the concatenated movable-strategy trajectory is evaluated using the same trajectory-level summary score as the static baselines,

$$F = -\alpha P_{\text{out}} + \beta P_{10} + \eta T_{\text{peer}}, \tag{24}$$

where P_{out} is the fraction of user snapshots below the configured SINR threshold, P_{10} is the trajectory-wide 10th percentile, and T_{peer} is a peer-weighted mean SINR term derived from UE-to-UE CSI. The coefficients (α, β, η) correspond to the configured outage, percentile, and peer tie-break weights. This final score is used for reporting and mode comparison; the online relocation controller itself is driven by one of $J_w^{(1)}(\mathcal{S})$, $J_w^{(2)}(\mathcal{S})$, or $J_w^{(3)}(\mathcal{S})$, depending on the selected movable-strategy variant. The above formulation defines the relocation mechanisms used throughout the Rabot simulations.

IV. Simulation Setup

The Rabot outdoor scenario is defined in `scenarios/rabot.yaml`. The scene is derived from an OpenStreetMap (OSM) boundary around the KU Leuven Gent Campus Rabot environment. Candidate distributed-AP positions are generated along building walls at low height, one rooftop proxy is reserved for the central reference strategy, and user mobility is simulated over the outdoor walkable area with mixed pedestrian and bicycle profiles.

The simulation workflow used by the paper matches the repository workflow:

1) build the outdoor scene and walk graph from OSM,
2) generate wall-mounted candidate APs and the single rooftop proxy site,
3) simulate user trajectories over the walkable outdoor area,
4) compute path-based CSI and explicit SINR terms,
5) compare the central proxy, the fixed distributed baseline, the historical local-CSI relocation rule, the nearest-user proxy-ESR rule, and the nearest-user proxy-power rule,
6) export reusable result files,
7) generate analysis figures in a separate postprocessing step.

This section documents the reproducible processing chain from scene generation to postprocessed figure export. The antenna configuration is kept fixed across all placement strategies at the element-model level: the simulator instantiates UPA-based node arrays with the 3GPP `tr38901` element pattern, half-wavelength spacing in both planar dimensions, and `VH` dual polarization. The distributed strategies use identical 2×3 wall-mounted APs, whereas the central reference aggregates that same element budget into one normalized co-located rooftop 6×6 array with summed transmit power. Consequently, the comparison isolates the effect of geometry, relocation, and distributed-versus-co-located topology while keeping the underlying element directivity model unchanged.

The figure set in Sections V to VII is read directly from the generated Rabot outputs under `outputs/rabot/` and `outputs/rabot/postprocessing/`. This keeps the manuscript synchronized with the latest postprocessed result files in the repository.

TABLE I: Key Rabot scenario parameters used in the Rabot simulations.

Parameter	Value
Carrier frequency	3.5 GHz
Bandwidth	100 MHz
AP array topology	2×3 UPA, dual-polarized `VH`
Central proxy topology	normalized co-located 6×6 UPA, dual-polarized `VH`
UE array topology	1×1 planar array, dual-polarized `VH`
Element spacing	0.5λ vertical, 0.5λ horizontal
Element radiation pattern	3GPP `tr38901`
User height	1.5 m
Number of users	10
Simulation horizon	3600 s
Time step	5 s
Distributed APs per strategy	6
Fixed distributed APs	0
Candidate wall height	1.5 m
Candidate wall spacing	18 m
Minimum candidate spacing	12 m
Candidate wall offset	0.25 m
Relocation window	60 s
Historical CSI decay rate	0.001 16 s^{-1}
Optimization-1 neighborhood size	$K = 10$ nearest UE snapshots per candidate
Optimization-2 distance threshold	25 m
Initialization seed	11
Compared strategies	central proxy, fixed distributed, movable distributed
Coverage maps	disabled in current Rabot config

TABLE II: Placement-strategy definitions used throughout the article.

Strategy	Description
`central_massive_mimo`	Single rooftop proxy on the building nearest the area center, oriented toward the center, with a normalized co-located 6×6 UPA obtained by aggregating the distributed element budget into one rooftop site. Use as central reference only.
`distributed_fixed`	One distributed subset is selected from wall candidates and kept unchanged for the full run.
`distributed_movable`	Optimization 1. The distributed subset is reselected every relocation window from the weighted 10th percentile of the stored best-user deployed-AP-to-UE SINR samples in the union of the configured $K = 10$ nearest UE-snapshot neighborhoods per candidate, with exponential aging of older window segments.
`distributed_movable_optimization_2`	Optimization 2. The distributed subset is reselected every relocation window from a nearest-user UE-to-UE proxy-CSI model: each candidate site binds to the closest user within the configured distance threshold, the code forms a wideband proxy channel from that user's measured UE-to-UE CFR, and the subset is ranked by proxy ESR.
`distributed_movable_optimization_3`	Optimization 3. The distributed subset is reselected every relocation window from the same nearest-user distance-threshold proxy model as optimization 2, but the code uses the stored UE-to-UE link-power row of the anchor user, replaces the diagonal entry by the strongest non-self value, rescales by $P_{\text{AP}}/P_{\text{UE}}$, and ranks the subset by proxy average power.

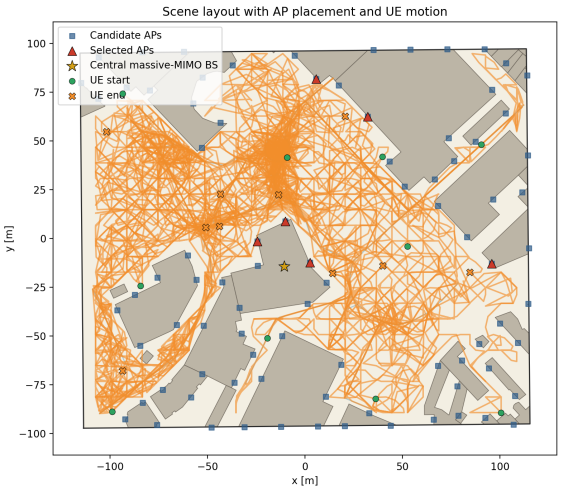
For the current Rabot instantiation, the relocation controller starts from a single seed constellation drawn from the wall-mounted candidate pool using the configured random seed. That seed constellation is frozen for the `distributed_fixed` baseline and also provides the initial labeled movable-AP positions for `distributed_movable`, `distributed_movable_optimization_2`,

and `distributed_movable_optimization_3`. Thereafter, every relocation window applies the greedy subset search from Section III with the selected optimization objective and then solves a minimum-displacement assignment so that schedule traces, relocation distances, and move fractions correspond to the motion of persistent AP identities rather than to arbitrary reordering of candidate sites. For optimization 2 and optimization 3, the dormant candidate-position AP-to-UE channel is never queried during this search; only measured UE-to-UE proxy information is used. Optimization 2 prefers the full complex wideband proxy channel and falls back to a power-based proxy-ESR computation when that proxy channel cannot be decoded robustly from the exported tensor. Optimization 3 always uses the stored proxy link-power tensor. In all cases, the actual deployed AP-to-UE CSI is recomputed only after the new candidate subset has been selected.

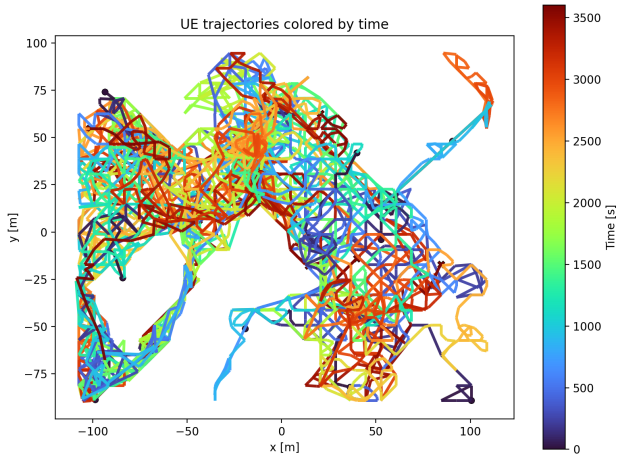
V. Scenario Geometry and Mobility Results

This result block summarizes the simulated environment before presenting radio-performance metrics. It reports the available candidate mounting locations, the selected deployment, and the spatial distribution of user trajectories over the Rabot area.

Figure 2 provides the geometric context for the subsequent SINR, ESR, and relocation analyses. The scene-layout panel reports the candidate-site pool and the selected constellation, whereas the trajectory colormap reports the spatial spread of the simulated user population. For the present Rabot instance, the wall-generation stage yields 87 candidate low-height mounting locations, from which six distributed sites are activated per distributed strategy. The simulated mobility trace contributes 7210 user-snapshot positions over the one-hour horizon, and these positions span most of the accessible outdoor footprint rather than a single narrow corridor. This geometry already suggests that a single rooftop proxy will face a harder coverage problem than wall-mounted distributed deployments, because the user population repeatedly moves through streets and courtyard edges that can be served from multiple nearby façades.



(a) Top-down Rabot scene layout with candidate sites, selected sites, and simulated mobility overlays.



(b) Trajectory colormap used to visualize the temporal spread of user motion through the walkable space.

Fig. 2: Geometry results for the Rabot scenario. The panels show the selected deployment in the urban layout and the temporal distribution of user motion over the walkable area.

VI. SINR- and ESR-Oriented Results

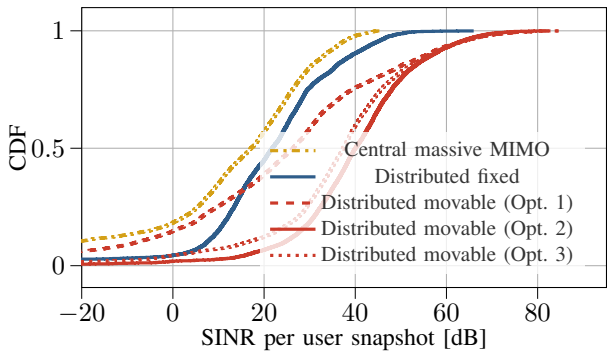
This section reports the radio-performance figures generated by the postprocessing workflow. The panels summarize global SINR statistics, user-level averages, temporal ESR evolution, aggregate ESR distributions, and worst-user behavior.

Figure 3 establishes a clear ordering. The nearest-user proxy-ESR controller of optimization 2 dominates almost the full empirical SINR range. Its mean SINR reaches 39.7 dB, with a 10th percentile of 24.5 dB; the fixed distributed baseline reaches 21.3 dB mean and 7.5 dB at the 10th percentile. The outage fraction below 0 dB drops from 4.41 % for the fixed baseline to 1.87 % for optimization 2. Optimization 3 is the second-best radio strategy with 35.7 dB mean SINR and 16.3 dB 10th-percentile SINR, but its outage fraction remains essentially identical to the fixed baseline at about 4.42 %. Hence, the power-only proxy improves the middle and upper portions of the distribution much more strongly than the weakest-link tail.

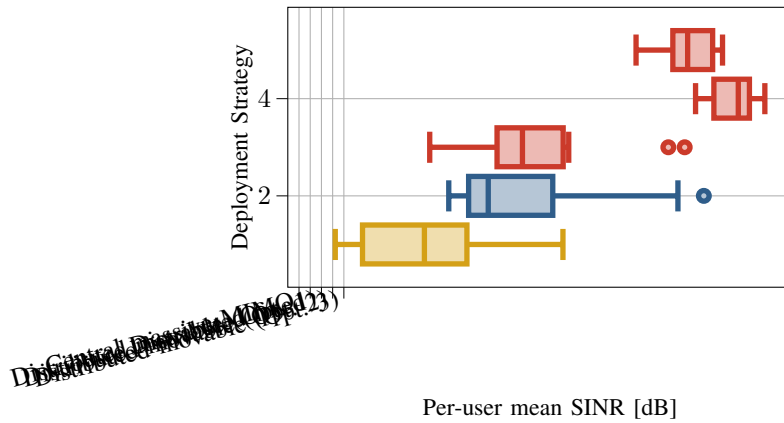
Optimization 1 behaves differently. Its mean SINR rises slightly above the fixed baseline to 22.8 dB, and it outperforms the fixed baseline in 44 of the 61 window-mean ESR values. Nevertheless, its 10th-percentile SINR collapses to −7.8 dB and its outage fraction rises to 14.7 %. The historical local-CSI rule therefore improves many typical snapshots while creating a much weaker lower tail. The central rooftop proxy is weakest throughout, with 9.8 dB mean SINR, −22.1 dB 10th-percentile SINR, and 18.4 % outage. The per-user mean boxplot reinforces the same message: optimization 2 lifts all users into a narrow 36.4 dB to 42.5 dB band, optimization 3 places all users in a lower but still clearly elevated 31.1 dB to 38.8 dB band, whereas the central proxy exhibits the widest user imbalance and even negative per-user means for the weakest users.

The ESR figures in Fig. 4 follow the same ranking. Mean ESR reaches 133.1 bit/s/Hz for optimization 2, 122.8 bit/s/Hz for optimization 3, 89.9 bit/s/Hz for optimization 1, 74.9 bit/s/Hz for the fixed distributed baseline, and 56.4 bit/s/Hz for the central rooftop proxy. Relative to the fixed baseline, optimization 2 improves the mean ESR by 77.8 % and the 10th-percentile ESR by 88.8 %. Optimization 3 still improves the mean ESR by 64.1 %, but remains consistently below optimization 2. Window by window, optimization 2 attains the highest mean ESR in 42 of the 61 relocation windows and exceeds the fixed baseline in all windows; optimization 3 leads in 15 windows and exceeds the fixed baseline in 57 windows; optimization 1 leads in only four windows. The relocation markers every 60 s make the controller action visible as a sequence of step changes, and the highest sustained post-relocation levels occur for optimization 2.

Together, these figures show that the best result in this scenario is not merely “more movement” or “more power”, but the combination of relocation with a phase-aware UE-to-UE proxy-CSI objective. The fixed distributed baseline remains a strong zero-relocation benchmark, whereas the central rooftop proxy is clearly mismatched to the spatially spread Rabot user process.

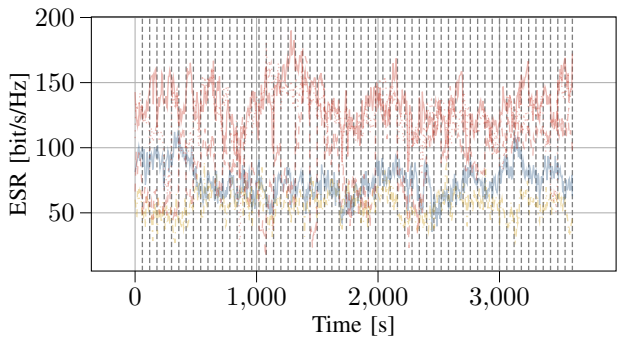


(a) Empirical SINR distribution across all user snapshots and compared placement strategies.

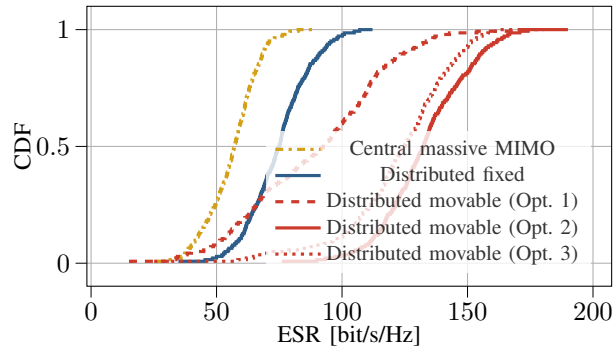


(b) Per-user mean SINR boxplot across strategies.

Fig. 3: First pair of radio-performance results. Optimization 2 dominates the global SINR distribution and shifts the full per-user mean-SINR boxplot upward, while the central rooftop proxy remains the weakest strategy.



(a) ESR over time, including relocation markers and window-level step behavior.



(b) ESR cumulative distribution function (CDF) across all evaluated snapshots.

Fig. 4: Second pair of radio-performance results. Optimization 2 produces the highest ESR levels over time and the most favorable global ESR distribution, with optimization 3 as the next-best movable strategy.

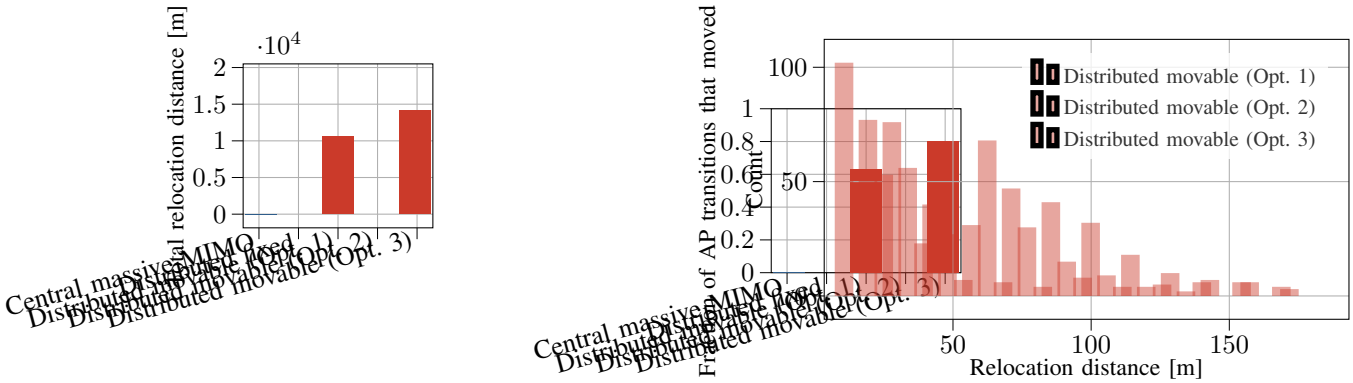
VII. Relocation and Practicality Results

The relocation figures complement the radio-performance results by quantifying the physical reconfiguration burden associated with movable infrastructure. The scheduling outputs report how often the movable APs change position and how much displacement the resulting schedules induce.

Together, the panels in Fig. 5 quantify the price paid for movable infrastructure. As expected, `central_massive_mimo` and `distributed_fixed` induce no physical motion. Among the movable controllers, optimization 1 is the least mobile: 228 of the 360 AP-transition events correspond to actual moves, for a move fraction of 63.3 %, a total traveled distance of 10.6 km, and a mean move distance of 46.5 m. Optimization 2 raises the move fraction to 80.0 % and the total traveled distance to 14.2 km, with a similar mean per-move distance of 49.2 m. Optimization 3 is the most aggressive controller, with 311 moves, a move fraction of 86.4 %, a total traveled distance of 18.6 km, and a mean move distance of 59.9 m.

This ranking is important when combined with the radio metrics from Section VI. Optimization 2 not only outperforms optimization 3 in mean and lower-tail SINR/ESR, it also requires less motion: about 24 % less total relocation distance and a smaller move fraction. Thus, within the two nearest-user proxy controllers, the phase-aware proxy-ESR rule is strictly preferable to the power-only proxy rule in the present scenario. Compared with optimization 2, optimization 1 reduces movement but gives up most of the radio-performance gain and substantially worsens the lower-tail robustness. The fixed distributed baseline therefore remains the practical zero-mobility reference, while optimization 2 defines the best radio-performance point among the movable strategies.

The histogram panel adds one further observation: the movable controllers do not differ only in how often they move, but also in how broadly they explore the candidate pool. Optimization 1 uses 131 distinct positions over the run, optimization 2 uses 140, and optimization 3 uses 162. Hence, the power-only rule is not merely more active; it is also spatially less conservative. In this dataset, that additional exploration does not translate into better radio performance.



(a) Strategy-level overview of total relocation distance and move fraction.

(b) Distribution of relocation distances across observed AP moves.

Fig. 5: Relocation-cost figures used to interpret how much physical infrastructure movement is required by the compared placement strategies. Optimization 2 achieves the strongest radio performance with a lower movement burden than optimization 3, whereas the fixed and central references remain static.

VIII. Discussion and Current Limits

The present study has several modeling and evaluation limits that define the scope of the reported results.

Three findings stand out from the Rabot results. First, distributed wall-mounted deployments are consistently superior to the normalized central rooftop proxy in this scenario. Even the static distributed baseline improves both average and lower-tail performance over `central_massive_mimo`, which indicates that the spatially spread outdoor user process benefits strongly from nearby façade-mounted service points. Second, the quality of the relocation proxy matters more than relocation alone. Optimization 2 outperforms optimization 3 on all reported radio metrics while also requiring less movement, which shows that phase-aware UE-to-UE proxy CSI carries useful information beyond power-only proxy measurements. Third, the optimization objective matters as much as the proxy source. Optimization 1 often improves window-average ESR, but it still produces a markedly worse lower tail than the fixed baseline. In the present dataset, local historical best-user SINR optimization is therefore not sufficient to control trajectory-level outage robustness.

The main limitations are the following:

- Only a single outdoor scenario is considered.
- The `central_massive_mimo` comparison point is a rooftop proxy, not a full placement model for one true co-located massive-multiple-input multiple-output (MIMO) array.
- The current Rabot configuration disables coverage-map exports, so the paper centers on trajectory-level SINR and schedule metrics.
- Citizen participation is abstracted as movable candidate positions and scheduler actions rather than being validated experimentally.
- Sensitivity to random seeds, scenario perturbations, and broader deployment-cost models is not yet characterized.

These limitations motivate extensions toward multi-scenario studies, sensitivity analyses, coverage-map reporting, and eventual experimental validation in co-created deployments.

IX. Conclusion

This article presented a simulation-based study of low-height distributed-MIMO access networks with repositionable infrastructure in an outdoor urban scenario. The paper combined a system model, three relocation algorithms, a reproducible Rabot simulation workflow, and postprocessed geometry, SINR, ESR, and relocation figures derived from exported artifacts. The three movable strategies span a historical local deployed-AP-to-UE controller that scores exponentially aged best-user SINR samples, a nearest-user UE-to-UE proxy-CSI controller with a proxy-ESR objective, and a nearest-user proxy-power controller that ranks subsets from stored UE-to-UE link-power samples scaled to the AP transmit-power budget.

For the Rabot dataset, the nearest-user proxy-ESR controller of optimization 2 is the best overall strategy. It yields the highest mean and lower-tail SINR/ESR, reduces outage most strongly, and does so with less movement than the power-only optimization 3 controller. Optimization 3 remains clearly beneficial relative to the fixed distributed baseline in mean-rate terms, but it does not improve the outage tail and requires the largest relocation effort. Optimization 1 improves average ESR relative to the fixed baseline, yet its much weaker lower tail shows that local historical best-user optimization does not automatically translate into robust trajectory-wide service. The fixed distributed baseline remains the strongest zero-relocation option, while the normalized central rooftop proxy is the weakest reference in this scenario.

Within its current scope, the framework supports systematic comparison of a central rooftop proxy, a fixed distributed deployment, and three relocation-capable distributed deployments under a common channel model and mobility process. Natural next steps are multi-scenario studies, coverage-map analysis, sensitivity analysis for the proxy-based relocation

rules, and experimental validation in the broader Cocoon program.